RooseBERT: A New Deal For Political Language Modelling - Supplementary Materials

Deborah Dore
Université Côte d'Azur
CNRS, INRIA, I3S
Sophia Antipolis, France
deborah.dore@univ-cotedazur.fr

Elena Cabrio
Université Côte d'Azur
CNRS, INRIA, I3S
Sophia Antipolis, France
elena.cabrio@univ-cotedazur.fr

Serena Villata
Université Côte d'Azur
CNRS, INRIA, I3S
Sophia Antipolis, France
serena.villata@cnrs.fr

***Abstract*—The increasing amount of political debates and politics-related discussions calls for the definition of novel computational methods to automatically analyse such content with the final goal of lightening up political deliberation to citizens. However, the specificity of the political language and the argumentative form of these debates (employing hidden communication strategies and leveraging implicit arguments) make this task very challenging, even for current general-purpose pre-trained Language Models (LMs). To address this, we introduce a novel pre-trained LM for political discourse language called RooseBERT. Pre-training a LM on a specialised domain presents different technical and linguistic challenges, requiring extensive computational resources and large-scale data. RooseBERT has been trained on large political debate and speech corpora (11GB) in English. To evaluate its performances, we fine-tuned it on multiple downstream tasks related to political debate analysis, i.e., stance detection, sentiment analysis, argument component detection and classification, argument relation prediction and classification, policy classification, named entity recognition (NER). Our results show improvements over general-purpose LMs on the majority of these tasks, highlighting how domain-specific pre-training enhances performance in political debate analysis. We release RooseBERT for the research community[1].**

***Index Terms*—language model, political debates, argument mining**

I. Training Hyper-Parameters

Table I shows the hyper-parameters for the best CONT and SCR RooseBERT models.

II. Additional Tasks

For the task of Motion Policy Classification, we tested also the Motion Policy Preference Corpus [1] (gold version). However, all models performed below random as shown in Table II. Therefore, we decided not to report these results in the main paper. The low performance is likely due to the fact that, from the original corpus, we only retained the gold labels (i.e., annotations produced by human annotators), discarding those generated by LLMs, which inevitably reduced the size of the corpus. This choice was motivated by consistency: since all other datasets in our benchmark rely exclusively on human annotations, retaining only the gold labels ensures a fair and uniform evaluation setting across the entire benchmark.

[1]huggingface.co/collections/MARIANNE-INRIA/roosebert

TABLE I
HYPER-PARAMETERS USED TO PRE-TRAIN ROOSEBERT WITH THE *base* ARCHITECTURE SETTING.

Hyper-parameters	**RooseBERT-cont**	**RooseBERT-scr**
Learning Rate Decay	Linear	Linear
Warmup Steps	10,000	10,000
Total Training Steps	150K steps	250K steps
Total Batch Size	2048	2048
Learning Rate	$3e^{-4}$	$3e^{-4}$
Weight Decay	0.01	0.01
Adam β_1, β_2, ϵ	0.9, 0.98, $1e^{-6}$	0.9, 0.98, $1e^{-6}$
Vocabulary Type	BERT	Custom
Vocabulary Size	28,997 (*cased*) 30,523 (*uncased*)	28,996 (*cased*) 30,522 (*uncased*)

TABLE II
AVERAGE ± STANDARD DEVIATION ON THE MOTION POLICY CLASSIFICATION TASK. RESULTS ARE EVALUATED USING MACRO F1 SCORE. STATISTICAL SIGNIFICANCE: ** $p < 0.05$ W.R.T. THE BEST MODEL (IN **BOLD**).

Models	**Motion Policy Classification**
BERT-base-cased	0.19 ± 0.20**
BERT-base-uncased	0.25 ± 0.05**
ModernBERT-base	0.38 ± 0.04
ConfliBERT-cont-cased	**0.39 ± 0.04**
ConfliBERT-cont-uncased	0.38 ± 0.05
ConfliBERT-scr-cased	0.13 ± 0.18**
ConfliBERT-scr-uncased	0.11 ± 0.14**
PoliBERTweet	0.27 ± 0.02**
RooseBERT-cont-cased	0.37 ± 0.04
RooseBERT-cont-uncased	0.39 ± 0.04
RooseBERT-scr-cased	0.30 ± 0.06**
RooseBERT-scr-uncased	0.32 ± 0.04**

III. Additional Error Analysis

To further investigate the results of RooseBERT, we plotted the confusion matrix of the best RooseBERT model against the confusion matrix of competitor models. This comparison allows us to understand in which scenarios RooseBERT improves over the other pre-trained models.

We chose four tasks: *sentiment analysis with HanDeSet* since it is the binary task in which RooseBERT gains the most w.r.t. BERT, *stance detection with AusHansard* to investigate the cross-domain adaptation abilities of RooseBERT w.r.t BERT, *Argument Component Relation Prediction and Classification with ElecDeb60to20* to understand why the results of RooseBERT are not statistically significant compared to PoliBERTweet and *Motion Policy Preference Classification with ParlVote+* to compare the results of RooseBERT in a classification setting with > 30 labels w.r.t. BERT.

A. Sentiment Analysis

Figure 1 shows the confusion matrix of RooseBERT-cont-cased against the confusion matrix of BERT-base-cased.

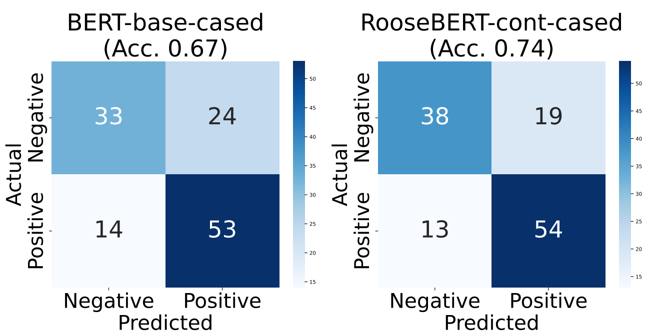

Fig. 1. Confusion Matrix of BERT and RooseBERT on the sentiment analysis task with dataset HanDeSeT.

In this task, both models achieved the best results using the *cased* configuration. RooseBERT is better at detecting the negatives (38 vs 33 correct), while positive detection remains nearly identical (54 vs 53). BERT tends to predict more positives, misclassifying 24 negative instances. RooseBERT reduces this to 19, suggesting it has learned a more precise representation of what constitutes a negative sentiment in parliamentary debate.

B. Stance Detection

Figure 2 shows the confusion matrix of RooseBERT-cont-cased against the confusion matrix of BERT-base-uncased.

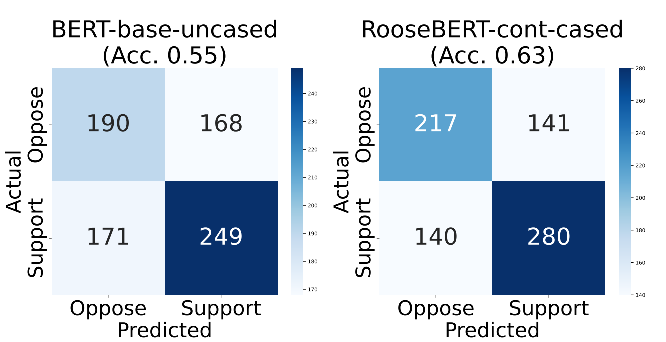

Fig. 2. Confusion Matrix of BERT and RooseBERT on the stance detection task with dataset AusHansard.

In this task, RooseBERT improves from 0.55 to 0.63, a meaningful gain in a cross-domain setting where generalisation is harder. RooseBERT improves on both classes symmetrically, better oppose detection (217 vs 190) and better support detection (280 vs 249), while also reducing errors in both directions (141 vs 168 false supports, 140 vs 171 false opposes). This is a notably balanced improvement, suggesting RooseBERT has learned discourse patterns that generalise across parliamentary styles.

C. Argument Component Relation Prediction and Classification

In this task, we decided to compare RooseBERT against PoliBERTweet to understand in which cases PoliBERTweet improves over RooseBERT and vice-versa. Figure 3 shows the confusion matrix of RooseBERT-scr-cased against the confusion matrix of PoliBERTweet.

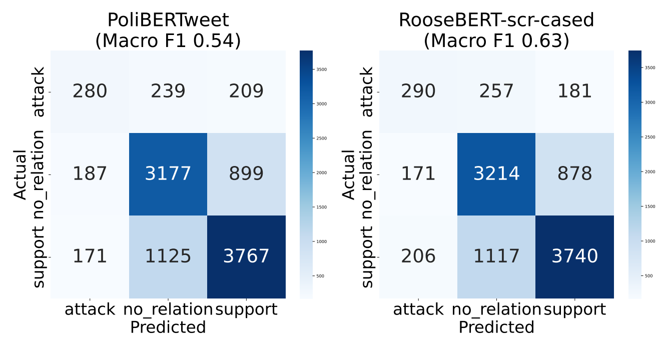

Fig. 3. Confusion Matrix of BERT and RooseBERT on the argument component relation prediction and classification task with dataset ElecDeb60to20.

Both models struggle with the *attack* class, which is the minority and hardest class, though RooseBERT improves slightly (290 vs 280 correct). On the other side, performance on *no_relation* and *support* is comparable across models: PoliBERTweet better classifies the support relation while RooseBERT is better at predicting the absence of a relation.

D. Motion Policy Classification

Figure 4 shows the confusion matrix of RooseBERT-cont-cased against the confusion matrix of BERT-base-uncased.

Overall, both models are generally predicting the correct policy category. Categories like European Union Positive, Political Authority: Party, Welfare State Expansion, Law and Order Positive, and Environmental Protection show clear diagonal concentration in both models, suggesting these are linguistically distinctive enough for both models to learn reliably. Many of the lighter rows indicate categories with very few correct predictions, particularly rare or conceptually similar classes like Constitutionalism Negative, Decentralisation Positive, Centralisation Positive, Education Limitation, and Immigration Positive. The diagonal appears slightly more concentrated for RooseBERT on several mid-frequency categories, suggesting it better discriminates between semantically similar policy pairs like Welfare State Expansion/Limitation and Immigration Positive/Negative.

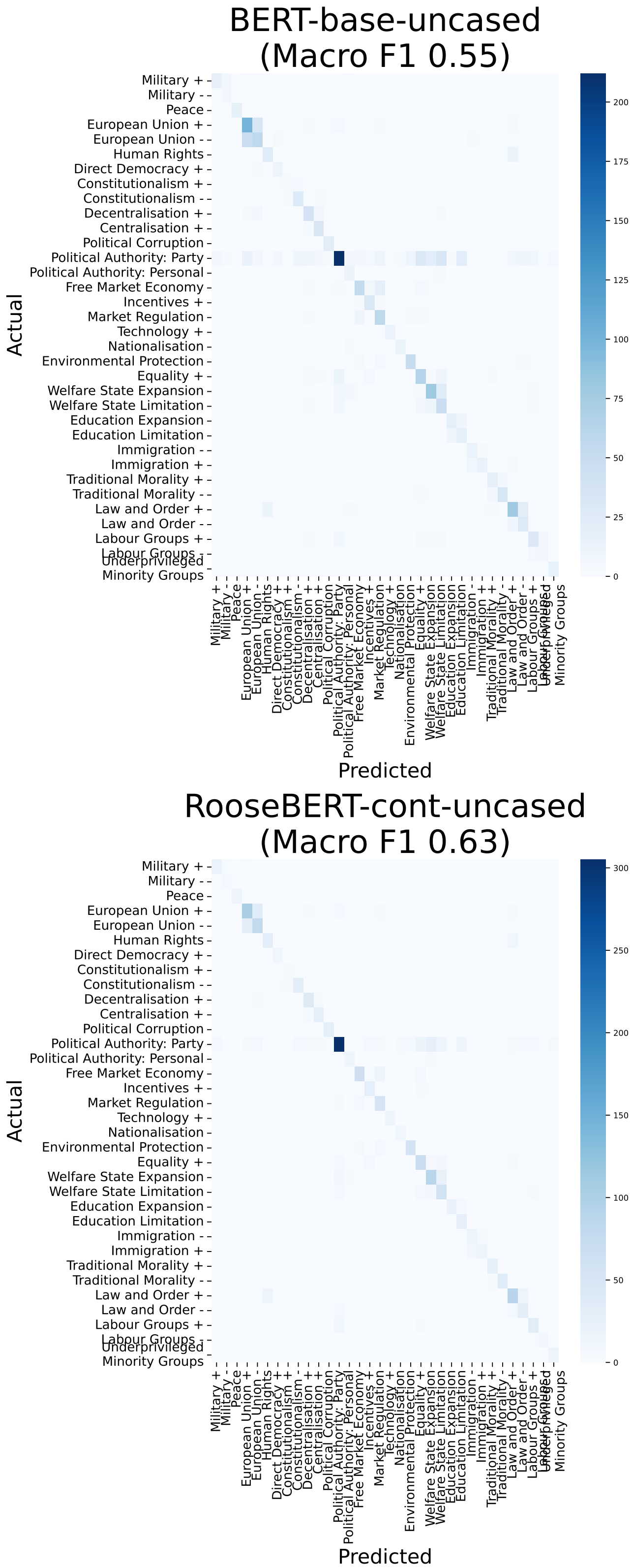

Fig. 4. Confusion Matrix of BERT and RooseBERT on the motion policy classification task with dataset ParlVotePlus.

IV. Experiments with LLMs

Large instruction-tuned language models (LLMs) such as LLaMA-3 [2], GPT4 [3], Mistral 7B [4] or Gemma 3 [5] have recently been explored for political discourse analysis through prompting strategies and few-shot learning ([6], [7]). While full fine-tuning of these models is often computationally prohibitive and unsuitable for resource-constrained environments, zero-shot and few-shot prompting make them a more practical choice for tasks such as sentiment analysis and argument component detection and classification. However, even with prompting, LLMs struggle to capture domain-specific features of political debates, such as implicit argumentation and turn-taking structures.

Table III reports the results for *gemma-3-4b-it*, *Mistral-7B-Instruct-v0.3*, *Llama-3.1-8B-Instruct*, and *gpt-4.1-nano* in the downstream tasks RooseBERT was tested for. All models were tested with a temperature of 0.0 to ensure deterministic outputs.

In Table III we see that the LLMs are mostly able to perform classification when fine-tuned, in one out of ten tasks surpassing RooseBERT (Argument Relation Prediction and Classification with the ElecDeb60to20 dataset). Overall, their performances are lower than RooseBERT's with results that are inversely proportional to the task's complexity: in argument component detection and classification and in motion policy classification particularly we see a deterioration of performances. On simpler tasks such as sentiment analysis and stance detection, these models are able to obtain comparable results under the fine-tuning setting.

Overall, these models are not exploitable for all kinds of tasks, their performances are tied to the prompt used and extensive training is required to achieve good results. Instead, RooseBERT can be fine-tuned on all specific tasks with high performances and it's tailored to political debates that require a specific focus nowadays.

A. LLMs Prompts

The following prompts were used when using the LLMs in zero-shot and few-shot settings:

Sentiment Analysis

You are a sentiment classification assistant. Classify the sentences using the following labels: positive, negative. Sentence: {sentence}. Output:

Stance Detection

You are a stance classification assistant. Classify the sentences using the following labels: support, oppose. Sentence: {sentence}. Output:

Argument Component Relation Prediction and Classification

You are a relation classification assistant. Classify the sentences separated by [SEP] using the labels: support, attack, no_relation. Sentence: {sentence}. Output:

Models	Sentiment Analysis		Stance Detection		Arg. Component Det. & Class.		Arg. Relation Pred. & Class.		Motion Policy Class.	NER
	ParlVote	**HanDeSeT**	**ConVote**	**AusHansard**	**ElecDeb**	**ArgUNSC**	**ElecDeb**	**ArgUNSC**	**ParlVote+**	**NEREx**
					one-shot					
gemma	0.44	0.53	0.49	0.44	0.21	0.23	0.37	0.30	0.02	0.01
Mistral	0.45	0.50	0.67	0.51	0.22	0.28	0.40	0.30	0.01	0.01
Llama	0.49	0.52	0.57	0.54	0.16	0.16	0.02	0.03	0.02	0.03
gpt	0.49	0.50	0.67	0.55	**0.28**	0.27	0.46	0.38	0.00	0.02
					3-shot					
gemma	0.44	0.44	0.63	0.53	0.21	0.23	0.27	0.33	0.07	0.02
Mistral	0.53	**0.57**	0.67	0.55	0.22	0.39	0.32	0.32	0.05	0.01
Llama	0.53	0.49	0.62	**0.57**	0.22	0.30	0.31	0.29	0.08	0.05
gpt	0.49	0.51	0.69	0.54	0.26	0.42	0.38	0.27	0.03	0.13
					fine-tuning					
gemma	0.61	**0.57**	0.68	0.49	0.08	0.53	0.63	0.43	0.26	0.71
Mistral	**0.79**	0.49	**0.75**	0.56	0.08	**0.57**	0.67	**0.65**	**0.47**	**0.85**
Llama	0.77	0.52	0.73	0.56	0.08	0.40	**0.68**	0.52	0.35	0.81

TABLE III

RESULTS OF LLMS ON DOWNSTREAM TASKS. BINARY TASKS ARE EVALUATED WITH ACCURACY; ALL OTHER TASKS USE MACRO F1. LLAMA = LLAMA-3.1-8B-INSTRUCT, MISTRAL = MISTRAL-7B-INSTRUCT-V0.3, GPT = GPT-4.1-NANO, GEMMA = GEMMA-3-4B-IT.

Argument Component Detection and Classification

Task: identify argumentative spans in the sentence. Span types:

- *¡claim¿: expresses a stance, opinion, or proposed policy*
- *¡premise¿: provides justification or support for a claim*

Rewrite the entire sentence exactly as given. Surround each identified span with ¡claim¿...¡/claim¿ or ¡premise¿...¡/premise¿. Only tag spans that clearly match one of the types. Do not add, remove, or change any words. Output the fully tagged sentence only. Sentence: {sentence} Output:

Motion Policy Classification

You are a motion policy preference classification assistant. Classify the motions using the following labels: Military: Positive, Military: Negative, Peace, European Union: Positive, European Union: Negative, Human Rights, Direct Democracy: Positive, Constitutionalism: Positive, Constitutionalism: Negative, Decentralisation: Positive, Centralisation: Positive, Political Corruption, Political Authority: Party, Political Authority: Personal, Free Market Economy, Incentives: Positive, Market Regulation, Technology: Positive, Nationalisation, Environmental Protection, Equality: Positive, Welfare State Expansion, Welfare State Limitation, Education Expansion, Education Limitation, Immigration: Negative, Immigration: Positive, Traditional Morality: Positive, Traditional Morality: Negative, Law and Order: Positive, Law and Order: Negative, Labour Groups: Positive, Labour Groups: Negative, Underprivileged Minority Groups. Sentence: {sentence}. Output:

NER

“Task: identify the entities in the sentence. Possible categories are: CC (coordinating conjunction, e.g. ’and’, ’but’), CD (cardinal number, e.g. ’three’), DT (determiner, e.g. ’the’), EX (existential ’there’), FW (foreign word, e.g. ’ad hoc’), IN (preposition or subordinating conjunction, e.g. ’because’), JJ (adjective, e.g. ’strong’), JJR (comparative adjective, e.g. ’stronger’), JJS (superlative adjective, e.g. ’strongest’), LS (list marker, e.g. ’1.’), MD (modal verb, e.g. ’must’), NN (singular noun, e.g. ’argument’), NNP (proper noun, e.g. ’Jennifer’), NNPS (plural proper noun, e.g. ’Americans’), NNS (plural noun, e.g. ’arguments’), PDT (predeterminer, e.g. ’all’), POS (possessive ending, e.g. ¨s¨), PRP (personal pronoun, e.g. ’she’), RB (adverb, e.g. ’clearly’), RBR (comparative adverb, e.g. ’more’), RBS (superlative adverb, e.g. ’most’), RP (particle, e.g. ’up’ in ’give up’), SYM (symbol, e.g. ’$’), TO (infinitive marker ’to’), UH (interjection, e.g. ’oh’), VB (base verb, e.g. ’argue’), VBD (past verb, e.g. ’argued’), VBG (gerund, e.g. ’arguing’), VBN (past participle, e.g. ’supported’), VBP (present verb, e.g. ’believe’), VBZ (3rd person present verb, e.g. ’believes’), WDT (wh-determiner, e.g. ’which’), WP (wh-pronoun, e.g. ’who’), , (comma), : (colon), . (dot), WRB (wh-adverb, e.g. ’why’). Rewrite the entire sentence exactly as given. Surround each identified entity span with ¡tag¿...¡/tag¿. Do not add, remove, or change any words. Output the fully tagged

sentence only. Sentence: $\{$*sentence*$\}$*. Output:*

References

[1] G. Abercrombie, F. Nanni, R. Batista-Navarro, and S. P. Ponzetto, "Policy preference detection in parliamentary debate motions," in *Proceedings of the 23rd Conference on Computational Natural Language Learning*. Association for Computational Linguistics, 2019, pp. 249–259.

[2] FacebookAI, "The llama 3 herd of models," 2024.

[3] OpenAI, "Gpt-4 technical report," 2024.

[4] M. AI, "Mistral 7b," 2023.

[5] G. Team, "Gemma 3 technical report," 2025.

[6] P. Törnberg, "Chatgpt-4 outperforms experts and crowd workers in annotating political twitter messages with zero-shot learning," *CoRR*, vol. abs/2304.06588, 2023.

[7] A. Kuila and S. Sarkar, "Deciphering political entity sentiment in news with large language models: Zero-shot and few-shot strategies," *CoRR*, vol. abs/2404.04361, 2024.